

PDF release of calc.xlsx

Sheet: Inputs

Parameter	Value	Unit	Source
Positive electrode thickness	7.56e-05	m	Chen2020 (unchanged)
Negative electrode thickness	8.52e-05	m	Chen2020 (unchanged)
Positive electrode porosity	0.45	—	r5_v11_final.json
Negative electrode porosity	0.42	—	r5_v11_final.json
Separator thickness / porosity	8e-6 / 0.60	m / —	r5_v11_final.json
Positive / negative particle radius	0.8e-6 / 0.8e-6	m	r5_v11_final.json
Electrolyte conductivity	1.8	S/m	r5_v11_final.json (constant, replaces Nyman2008 fn)
Electrolyte diffusivity	6e-10	m ² /s	r5_v11_final.json
Cation transference number	0.55	—	r5_v11_final.json
Current collectors Al / Cu	10e-6 / 6e-6	m	r5_v11_final.json
Total heat transfer coefficient	120	W/m ² /K	r5_v11_final.json
Nominal cell capacity	5.085	Ah	r5_v11_final.json (R5: = measured 1C cap)
Electrode area (height x width)	0.1027	m ²	Chen2020: 0.065 x 1.58
Voltage window	2.5 - 4.2	V	Chen2020
Positive max concentration	63104	mol/m ³	pybamm Chen2020
Negative max concentration	33133	mol/m ³	pybamm Chen2020

Sheet: CapacityEnergy

Quantity	Value	Unit	Formula	Source
1C discharge capacity	5.084876616106298	Ah	CC 5.085 A to 2.5 V event	r5_v11_1c_spme.json:capacity_ah
4C charge accepted	0.9197 C-h x 5.085 = 4.6767	Ah	charge capacity_ah = duration x C/3600; true Ah = reported x nominal	r5_v11_4c.json:capacity_ah
4C acceptance fraction	92.0	% of 1C capacity	accepted / 1C capacity	mechanical derivation
Discharge energy (1C)	18.62217418355077	Wh	integral V*I dt	r5_v11_energy.json:energy_wh
Midpoint voltage	4.1299	V	energy / capacity	r5_v11_energy.json:midpoint_voltage_v

Sheet: EnergyDensity

Quantity	Value	Unit	Formula	Source
Energy density (mass)	605.3345216842939	Wh/kg	18.6222 Wh / 0.0307634 kg	r5_v11_energy.json
Energy density (volume)	981.200981695033	Wh/L	18.6222 Wh / 1.8979e-5 m3	r5_v11_energy.json
Criterion	>= 392.61	Wh/kg	entry 0 stage2	log.jsonl entry 0
Verdict	PASS (605.33 >= 392.61)	—	mechanical compare	log-evaluate R5

Sheet: NP_Mass

Layer	Mass (g)	Formula	Source
Positive electrode	13.9296	$75.6\mu\text{m} \times 0.55 \times 3262 \times 0.1027$	r5_v11_energy.json:layer_kg_m2
Negative electrode	8.4093	$85.2\mu\text{m} \times 0.58 \times 1657 \times 0.1027$	r5_v11_energy.json:layer_kg_m2
Positive CC (Al)	2.7729	$10\mu\text{m} \times 2700 \times 0.1027$	r5_v11_energy.json:layer_kg_m2
Negative CC (Cu)	5.5212	$6\mu\text{m} \times 8960 \times 0.1027$	r5_v11_energy.json:layer_kg_m2
Separator	0.1305	$8\mu\text{m} \times 0.4 \times 397 \times 0.1027$	r5_v11_energy.json:layer_kg_m2
Total	30.7634	sum	r5_v11_energy.json:mass_kg
Geometric N/P	0.624	$(0.58 \times 33133 \times 85.2\mu\text{m}) / (0.55 \times 63104 \times 75.6\mu\text{m})$	parameter-set keys; caveat: Chen2020 neg OCP saturates above stoich 0.90 -> practical N/P verified by plating-free 4C sim instead

Sheet: Process

Parameter	Value	Unit	Formula	Source
Positive areal density	135.63	g/m ²	$75.6\mu\text{m} \times 0.55 \times 3262$	layer_kg_m2 x 1000
Negative areal density	81.88	g/m ²	$85.2\mu\text{m} \times 0.58 \times 1657$	layer_kg_m2 x 1000
Positive compaction density	1.79	g/cm ³	$3262 \times 0.55 / 1000$	derived (note: /1000)
Negative compaction density	0.96	g/cm ³	$1657 \times 0.58 / 1000$	derived (note: /1000)
Electrolyte fill amount	9.19	g	$7.662\text{ cm}^3 \times 1.2\text{ g/cm}^3 \times 1.0\text{ fill}$	literature density 1.2, annotated
Formation recommendation	0.1C CC to 4.2V, 25C, 2 cycles	—	design recommended value; production tuning required	annotated